RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 5 (Delta)
Topic(s): Vectors
(Vectors II and III (Questions))

1 Referred to the origin O, points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively such that angle $AOB = 30°$ and $|\mathbf{a}| = 4|\mathbf{b}|$. Point D is the mid-point of OA and point E lies on OB such that $OE : EB = 3 : 2$.

(i) Find the position vectors $\overrightarrow{OD}$ and $\overrightarrow{OE}$, giving your answers in terms of $\mathbf{a}$ and $\mathbf{b}$. [2]

(ii) Show that the vector equation of the line BD can be written as $\mathbf{r} = \lambda\mathbf{a} + (1-2\lambda)\mathbf{b}$, where λ is a parameter. By finding the vector equation of the line AE in a similar form in terms of a parameter μ, find, in terms of $\mathbf{a}$ and $\mathbf{b}$, the position vector of the point F where the lines BD and AE meet. [4]

2 Planes p and q are perpendicular. Plane p has equation $\mathbf{r} \cdot (\mathbf{i} + 2\mathbf{j} - \mathbf{k}) = 6$. Plane q contains the line l with equation $\mathbf{r} = \mathbf{j} - 12\mathbf{k} + \lambda(4\mathbf{i} + \mathbf{j} + 10\mathbf{k})$ where λ is a parameter.

(a) Find a cartesian equation of q. [2]
(b) Find a vector equation of the line m, where p and q meet. [2]

Point A has coordinates $(0, 1, -12)$. l passes through p at point B. Point C is the foot of the perpendicular of A to p.

(c) Find the position vector of B. [2]
(d) Find the position vector of C. [2]
(e) Find the exact area of triangle ABC. [2]

3 The plane p_1 contains the point $A(-4,4,0)$ and the line with equation $\frac{x-2}{2}=\frac{y-4}{3}, z=6.$

The plane p_2 has equation $\mathbf{r}=\begin{pmatrix}2\\4\\6\end{pmatrix}+\lambda\begin{pmatrix}1\\0\\-1\end{pmatrix}+\mu\begin{pmatrix}0\\1\\1\end{pmatrix}$, where λ and μ are real constants.

(a) Show that the line of intersection between p_1 and p_2 is parallel to the vector $\begin{pmatrix}5\\6\\1\end{pmatrix}$.

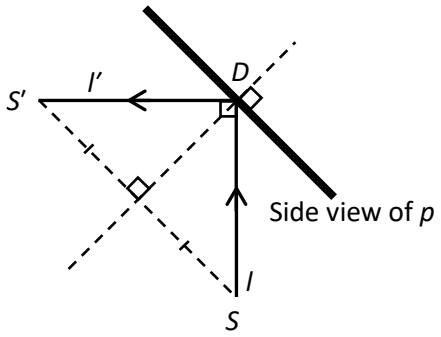
Hence write down the vector equation of the line of intersection between p_1 and p_2. [5]

(b) Determine the acute angle between p_1 and p_2. [2]

(c) Find the position vector of the foot of perpendicular from point A to p_2. [3]

(d) The plane p_3 has equation $ax+3y+2z=b,$ where $a,b\in\mathbb{R}$. Find the values of a and b such that all three planes have a common line of intersection. [2]

4



The diagram above shows a glass prism. The surface of the prism is part of the plane p with equation $by - z = 4$, where b is an integer constant. A ray of light l passes through a point S with coordinates $(5, 1, 3)$, and travels in a direction parallel to the unit vector $\frac{2}{3}\mathbf{i} + c\mathbf{j} + \frac{2}{3}\mathbf{k}$, where c is a negative constant, until it hits the surface of the prism at point D. The light ray is reflected by the surface of the prism. The reflected ray of light, l', passes through the point S', where S' is the image of point S. It is given that the lines l and l' are perpendicular and they lie on the same plane as the normal to p passing through point D. Also, the acute angle between l and the normal to p is equal to the acute angle between l' and the normal to p.

(i) Find the exact value of c and show that $b = 1$. [4]

(ii) Find the coordinates of point D. [3]

(iii) Show that the coordinates of S' are $(m, -1, 1)$ for some integer constant m to be determined and find a vector equation of l' in exact form. [5]

5 The line l_1 has equation $\mathbf{r} = 11\mathbf{i} - 3\mathbf{j} - \mathbf{k} + \lambda(\mathbf{i} - \mathbf{j} + 2\mathbf{k})$, where λ is a parameter. The point A has coordinates $(2, 0, -1)$.

(a) The plane p contains the line l_1 and the point A. Find a cartesian equation of the plane p.

[3]

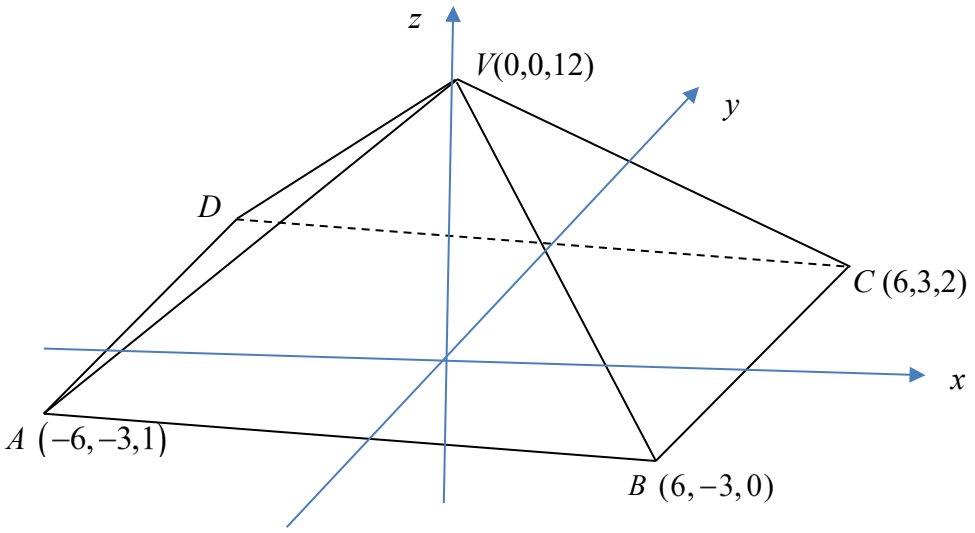
(b) Find the position vector of the point A', the reflection of the point A in the line l_1. [4]

The line l_2 passes through the point A, and is perpendicular to the line l_1. Planes p and q are perpendicular planes such that p and q meet at line l_2.

(c) Find a vector equation of the line l_2. [2]

(d) The plane $\prod$ is such that $\prod$ contains the point $(11, -3, -1)$ and is parallel to q. Find the perpendicular distance between $\prod$ and q. [2]

6



An oblique pyramid has a plane base $ABCD$ in the shape of a parallelogram. The coordinates of A, B and C are $(-6,-3,1)$, $(6,-3,0)$ and $(6,3,2)$ respectively. The apex of the pyramid is at $V\,(0,0,12)$.

(i) Find the coordinates of D. [1]

(ii) Find the cartesian equation of face BCV. [3]

(iii) Find the angle between face BCV and the base of the pyramid. [3]

(iv) Find the shortest distance from the midpoint of edge AD to plane BCV. [3]